Lesson Plan

Teacher Name: Ms. Lubna Kakkengal and Mr. Dilan Devendra	**Class/Section:** Grade 6
Subject: TCD: Photon AI Mission	**Date: Week 12 (16 Nov to 20 Nov)**
Class Description: Grade 6 TCD class, Photon AI Mission Session 1 of 2 (2 hours). Students explain the Photon robot as a sense-think-act system, label a system diagram, and write and test multi-step algorithms with turns at specified angles, updating a flowchart as they go.	
TA/LSA Role (if applicable):	

Differentiated Learning Groups (Data)			
Desert Fox	**Camel**	**Oryx**	**Falcon**

Lesson Overview		
Lesson Topic	Photon AI: Sense, Think, Act	
Standard(s)	Technology: algorithms, inputs and outputs; Math: 6.G (angles and distance), 6.EE (variables); Science: systems and cause and effect	
Learning Objective(s) (WALT): We are learning to describe a robot as a system and to program precise multi-step movements.		**Success Criteria (WILF):** I can explain sense, think and act using Photon as an example. I can label Photon's sensors (inputs) and outputs on a system diagram. I can program a route with distances and turns at set angles. I can draw a flowchart for my algorithm and update it when I change the code.
Literacy Strategies	System diagram labels; flowchart symbol cards (start or end, process, decision); frame: Photon senses ... thinks ... and acts by ...	
Assessment	Labelled system diagram; flowchart and tested route; live quiz on the TCD website.	
Targeted Vocabulary	system, input, output, sensor, algorithm, flowchart, sequence, angle, distance encoder	
Skills	Computational thinking, systems thinking, precision	

	documentation
National Identity Integration	Link to the UAE's AI strategy and robots used in Dubai, such as delivery robots and smart police stations, as real sense-think-act systems.
AI/Digital Learning	Photon robots and tablets with the Photon coding app; hook video; live quiz on the TCD website. Presentation: Grade6_Week12_Presentation.pptx. Worksheet: Grade6_Week12_Worksheet.

Lesson Structure	
Starter (Hook) Engage and Ask (15 minutes)	Warm-up: Sense, Think, Act Charades (slide 4). Show the hook video and ask the hook question (slide 2). Share the learning goals (slide 3) and key words (slide 5).
I Do (25 minutes)	Explain robots as systems (slide 6). Show Photon's sensors and outputs and label the diagram together (slide 7). Model coding a precise move and turn (slide 8) and drawing a flowchart (slide 9).
We Do (20 minutes)	Pairs label the system diagram. Class codes a short square together and compares the result with the plan (slide 10).
You Do (45 minutes)	Pairs plan, code, test and adjust their route and keep the flowchart up to date (slide 12). Challenge levels on the board (slide 13). Live quiz on tablets (slide 14).
Differentiated Instruction: **Beginning / On / Above**	**Beginning: Code a square route with 90 degree turns using the flowchart template.** **On: Code a route with at least three different turn angles.** **Above: Code a route that draws a regular hexagon and explain the angle calculation (360 divided by 6).**
Closure/ Reflection (15 minutes)	Pairs share how far off the target they were and what they adjusted. Exit ticket (slide 15). Robots back to charging.

Inclusive Planning
Students receiving additional inclusive support (IEP or ILPs) or Gifted and Talented: EAL: system diagram label cards and flowchart symbol cards. SEN: square route with template; mixed-ability pairs. G&T: hexagon with angle calculation.
Inclusion / EAL Support

☑ Visual Supports
☐ Chunked Instructions
☐ Reduced Language Load
☑ Model / Worked Example
☐ Pre-teach Vocabulary
☐ TA/LSA Support
☐ Extra Processing Time
☐ Text Read by TA/LSA
☑ Flexible Grouping
☐ Assistive Technology
☐ Movement Break
☑ Scaffold / Sentence Starters
☐ Other:

G&T / Extension

☑ Extension / Challenge Task
☑ Higher-order Questioning
☑ Depth & Complexity Task
☐ Independent Inquiry
☐ Leadership / Peer Coaching
☐ Real-World Application
☐ Curriculum Compacting
☐ Above-grade-level Texts
☐ Student-led / Passion Project
☐ Open-Ended Task
☐ Other:

Assessment Adaptations

☑ Oral Response
☐ Reduced Questions
☑ Scaffolded Success Criteria
☐ Extra Time
☐ Alternative Outcome Format
☑ Teacher Conferencing/Live Feedback
☐ Quiet / Separate Space
☐ Other: